

Adaptive RF-Visible Light Hybrid Offloading for Congestion Aware Indoor 6G Networks

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Abstract—Nowadays most of the internet usage happens inside classrooms, offices and buildings. Because of this, Wi-Fi networks often become slow when many users connect at the same time. This creates congestion in the RF spectrum, especially when students use mobile phones, laptops and online study materials together. Since 6G aims to provide very high speed and low delay handling indoor congestion is an important challenge. In this paper we studied a simple solution by combining Wi-Fi (RF) with Li-Fi (VLC) technology. Li-Fi uses LED lights to send data and can provide very high speed, but it works properly only when the device is fixed and under the light. So instead of using only Wi-Fi for all devices, we propose a hybrid system where stationary devices like desktop computers, smart boards and projectors are connected to Li-Fi while mobile devices continue using Wi-Fi. By doing so the load on Wi-Fi reduces during high attendance. Our classroom-based analysis shows that this method can improve overall speed, reduce congestion and make the network more efficient. This hybrid approach can be useful for future smart classrooms and indoor 6G environments.

Keywords—6G networks, hybrid RF-VLC, Li-Fi, congestion-aware systems, adaptive controller, indoor networks, energy efficiency.

I. INTRODUCTION

Wireless communication is developing very fast, and now researchers are already talking about 6G networks [1], [2], [3]. 6G is expected to provide very high data speeds, low latency, and support many connected devices at the same time. In real life, we can see that most internet usage happens indoors, especially in classrooms where students use mobile phones, laptops, online PDFs, videos, and smart boards during lectures.

The problem with Wi-Fi is that it works on limited RF spectrum [1], [10]. When many students connect at the same time, the bandwidth gets divided and the speed reduces. Sometimes there is delay while opening websites or streaming videos. As attendance increases, this congestion becomes more noticeable. So, depending only on RF communication may not be enough in future.

To reduce this problem researchers are exploring Optical wireless communication technologies. One of them is Li-Fi which uses LED light to transmit data [4], [14]. The visible light spectrum is much larger than RF spectrum and it can provide very high speed. However,

Li-Fi has one limitation it needs line of sight and works better for stationary devices. Hence, instead of replacing Wi-Fi completely combining both technologies makes more sense [8], [12].

It was observed that there are network congestion problems in the classrooms and college campus. During lectures or in library many students connect their mobile phones and laptops to the same Wi-Fi network at the same time. Sometimes while downloading notes, opening PDFs, or watching educational videos, the internet speed becomes slow and buffering issues occur. Hence, depending only on Wi-Fi isn't sufficient for future high-speed requirements. Therefore, in this paper a hybrid RF-Li-Fi system for classroom environments is proposed. In our idea mobile devices continue using Wi-Fi while fixed devices like projectors and desktops are shifted to Li-Fi during high load. This helps reduce RF congestion and improves overall network performance in a simple and practical way.

Since 6G is expected to provide ultra-high speed and support advanced applications like VR learning and smart classrooms, it can be explored as a practical solution that can improve indoor connectivity. New communication technologies, such as Li-Fi which uses LED lights to transmit data can be used because classrooms already use LED lighting, so combining lighting and communication seems like an efficient idea.

Another reason for choosing this topic is that hybrid RF-Li-Fi systems are being discussed for future networks, but many examples are theoretical. But how will this concept be applied in a simple classroom scenario that is experience daily needs to further studied. By analyzing attendance variation and device behavior, technical concepts to real life situations can be achieved.

This topic provides an opportunity to learn about congestion management, bandwidth calculation, and adaptive network control. Connecting theoretical 6g concepts with practical implementation ideas in educational environments can be explored further.

II. LITERATURE REVIEW

For 6G networks and indoor communication systems several research papers were referred to understand the current developments in this field. Many researchers have already discussed the challenges that future

wireless networks may face, especially regarding spectrum congestion and increasing data demand.

Ian F. Akyildiz and other researchers explained the roadmap towards 6G networks and highlighted that future networks will require ultra-high speed, latency and support for massive device connectivity. Their work mainly focused on advanced RF technologies and intelligent networking. However, they also mentioned that limited spectrum availability could become a serious problem as the number of connected devices keeps increasing [1]. M.Z. Chowdhury and colleagues further discussed the 6G wireless communication systems, outlining applications, requirements, and challenges for next-generation networks [2].

Harald Haas introduced the concept of Li-Fi and demonstrated that visible light communication technology can achieve very high data rates using LED light sources. His work proved that Li-Fi has strong potential as an alternative communication technology. But most of the early research focused on standalone Li-Fi systems and did not deeply discuss how it can work together with RF networks in real time environments [4], [11]. Reports from the International Telecommunication Union (ITU) on IMT-2030 vision also emphasize that future 6G systems may require integration of multiple technologies instead of relying only on RF communication. This shows that hybrid communication systems could be an important direction for future research [3]. Some researchers have also studied hybrid RF-VLC systems and basic load balancing methods. In many of these studies, user allocation was done using fixed rules or simple threshold values. Although these methods help in reducing congestion to some extent, they do not fully consider real-time classroom behavior, device types or dynamic traffic changes [8], [9], [12].

From reviewing these studies, it was observed that while hybrid systems are discussed, there is still scope for improving adaptability and practical indoor implementation. Most papers focus on theoretical models or simulations, but fewer works explain simple real life classroom scenarios. Therefore, this paper aims to design a congestion aware hybrid RF-Li-Fi system specifically for a classroom environment considering attendance variation, device behavior, and realistic bandwidth estimation.

III. PROBLEM STATEMENT

The indoor 6G network has these following problems:

RF Spectrum Congestion -Limited bandwidth cannot sustain increasing indoor traffic demand.

High-Bandwidth Applications-VR, AR, and industrial automation require multi-gigabit speeds.

Dynamic User Behavior -Stationary and mobile users require differentiated connectivity.

Energy Consumption-Increased RF transmission raises operational power usage.

Limited Adaptivity- Existing hybrid systems lack intelligent real-time switching mechanisms.

Objective:

Develop a congestion-aware hybrid RF-VLC system capable of dynamically offloading stationary users to VLC, reducing RF load and improving network efficiency.

IV. PROPOSED HYBRID DESIGN

The proposed design consists of components: 6G RF Base Station, Indoor RF Access Points, Li-Fi LED Access Points and an Adaptive controller. The Adaptive controller includes a Congestion Monitor which will monitor RF usage, a Traffic Allocator which will allocate stationery to high bandwidth users, a Handover Manager which will execute a smooth RF to VLC switching.

Consider a smart classroom of area 200 to 220m² which has strength of 63 students, a faculty member. The class has a stationary projector, a smartboard and a desktop computer, all these equipment are used during the lectures. The students attendance varies daily.

In the proposed design, the student's mobile phones and laptops are connected to RF (Wi-Fi) because these devices are not stationary during the lectures. Li-Fi requires stable line of sight conditions; it is more suitable for fixed devices. Therefore, the desktop computer, projector, and smart board are connected to Li-Fi to reduce RF congestion and improve bandwidth distribution.

To understand the system's behavior, classroom attendance is assumed under three different conditions: low, medium and high.

Assumptions Used:

Classroom Wi-Fi capacity: 300 Mbps

Bandwidth per student device: 5 Mbps

Stationary devices:

Desktop: 10 Mbps

Smart board: 12 Mbps

Projector: 8 Mbps

Total Stationary traffic: $10 + 12 + 8 = 30$ Mbps

- In the Low Attendance Conditions:

Students present: 28

Total RF Traffic: $28 * 5 = 140$ Mbps

RF Utilization: $(140/300) * 100 = 46.6\%$

Approximate RF usage: 40% to 50%

Under this condition the RF network will be able to handle traffic. There will be minimal delays and stable internet access for the students. Li-Fi supports only stationary devices, but congestion is not significant in this condition.

- In the Medium Attendance Condition:

Students present: 42

Total RF traffic: $42 * 5 = 210$ Mbps

RF Utilization: $(210/300) * 100 = 70\%$

Approximate RF usage: 65% to 75%

During this condition, more students will access more online notes, PDFs, etc. The load on RF increases and slight latency may be observed. However, since high bandwidth fixed devices will be shifted to Li-Fi, the overall network performance will remain stable.

- In the High Attendance Condition (Without Li-Fi Offloading):

Students present: 52

Total RF traffic: $52 * 5 = 260$ Mbps

RF Utilization: $(260/300) * 100 = 86.6\%$

Approximate RF Usage: 85% to 90%

In high attendance condition without Li-Fi Offloading, RF usage climbs to 85% to 90%, creating heavy congestion. This leads to noticeable delays and unstable connectivity, making the network less reliable for students.

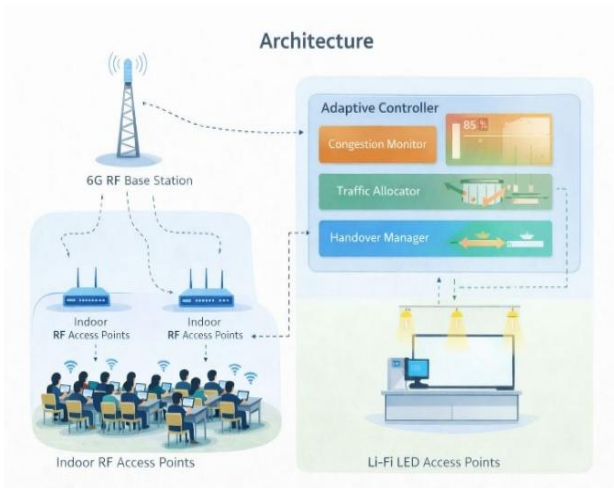


Fig.1. Design of the proposed system.

Peak Attendance Condition

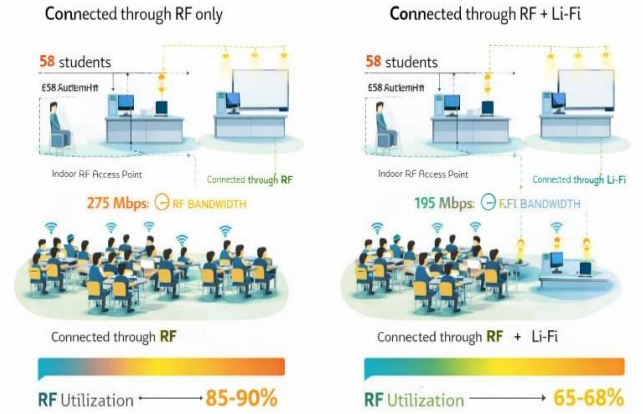


Fig.4. High attendance conditions

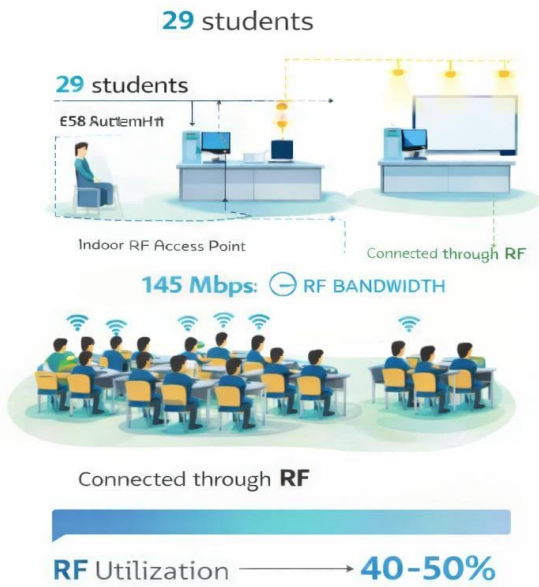


Fig.2. Low attendance conditions.

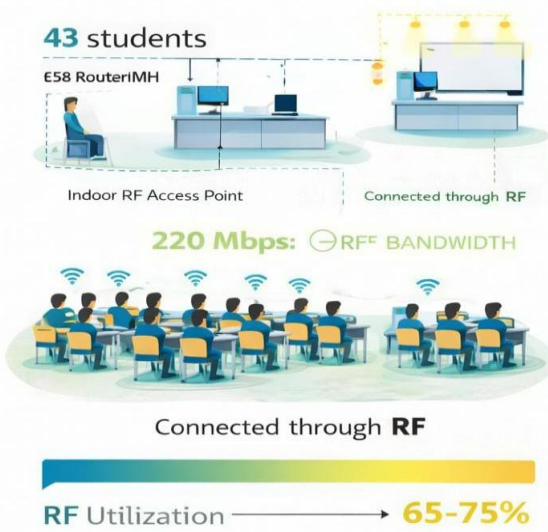


Fig.3. Medium attendance condition

- In the High Attendance Condition (With Li-Fi Offloading):

Stationary Traffic (30 Mbps) shifted to Li-Fi, therefore
New RF Traffic: $260 - 30 = 230$

RF Utilization: $(230/100) * 100 = 76.6\%$

However, in real classrooms not every student uses internet, assuming 15% idle traffic reduction.

RF Utilization: $76.6 * 0.85 = 65.1\%$

Approximate RF Usage: 65% to 68%

In high attendance conditions with Li-Fi Offloading, RF usage settles around 65% to 68%. This keeps the network balanced, ensuring smooth performance and stable connectivity with only minimal delays.

This estimation is based on analytical bandwidth calculation. Assuming an RF capacity of approximately 300 Mbps and an average usage of 5 Mbps per student, then the total demand at full attendance becomes:

$63 * 5 \text{ Mbps} \sim 315 \text{ Mbps}$

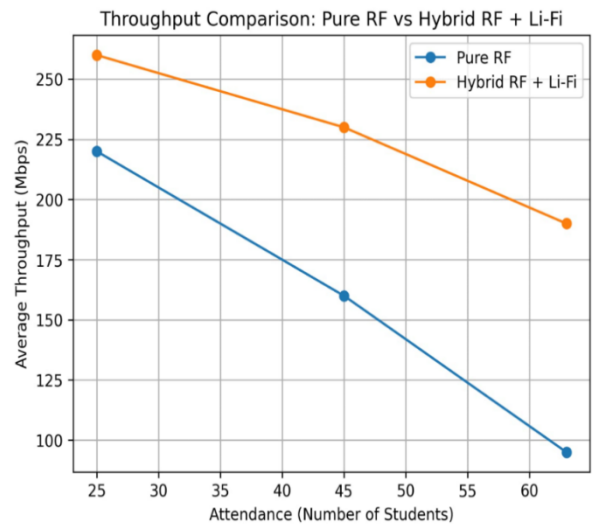


Fig.5. Throughput improvement due to hybrid system.

Since the demand exceeds available capacity, congestion increases. By shifting fixed devices such as the desktop, smart board, and the projector (approximately 30 Mbps) to Li-Fi, the load on RF reduces to nearly 65% to 68% improving the overall performance in the classroom environment.

V. METHODOLOGY

In this work a classroom based analytical approach was used to study the performance of the hybrid RF-Li-Fi system. First, it was assumed that there was an indoor classroom scenario with varying attendance levels. The total number of active devices during low, medium and high attendance was noted down. An average bandwidth requirement per student device was considered based on activities such as video streaming and downloading study material we assumed an approximate RF capacity and calculated the total bandwidth demand under each condition. Then, analysis was carried out on how shifting stationary devices like desktop, smart board, and projector to Li-Fi reduces the load on RF. RF utilization before and after offloading was compared to observe performance improvement. Based on these calculations and comparisons, the effectiveness of the proposed hybrid model was evaluated.

VI. ADVANTAGES OF HYBRID SYSTEM

- Reduced RF congestion
- Improved spectral efficiency
- Higher per user throughput
- Enhanced indoor security
- Energy efficient dual use LED lighting

VII. LIMITATIONS

- Line of sight dependency
- Infrastructure installation cost
- Limited mobility support under VLC

VIII. Li-Fi VS Wi-Fi COMPARISON

Table 1: RF vs. Hybrid System Performance Metrics.

Performance Comparison Between RF and the Hybrid System		
Parameter	Parameter	Parameter
RF Utilization	RF Utilization	RF Utilization
Avg Throughput	Avg Throughput	Avg Throughput
Packet Loss	Packet Loss	Packet Loss

Table 2: Comparative Features of Wi-Fi and Li-Fi.

Li-Fi vs Wi-Fi Comparison		
Feature	Wi-Fi	Li-Fi
Speed	Moderate	Very high
Coverage	Throughput	Line of sight
Mobility	High	Limited
Security	Moderate	High
Interference	Susceptible	Minimal

IX. REAL WORLD APPLICATIONS

Hybrid RF-VLC systems can be applied in:

- Smart offices: Desktops and mobile laptops supported efficiently.
- Hospitals (EMI free zone): Secure patient data transfer with reduced RF interference.
- Airports: High density users requiring reliable indoor

connectivity.

- VR/AR laboratories: Ultra high throughput for stationary VR stations.
- Industrial automation.
- Smart homes: Secure and high-speed data for IoT devices.
- Smart Classrooms: High speed data for VR learning and stationary tablets.

X. SECURITY AND ENERGY EFFICIENCY

Li-Fi signals don't penetrate walls, improving physical layer security. Reduced RF transmission decreases interference and energy consumption. Since LEDs serve dual purposes (lighting and communication), overall power utilization improves.

XI. CONCLUSION

The hybrid RF-Li-Fi system in the classroom scenario has a strong potential to improve indoor network performance in real educational environments. Classrooms today have a mix of stationary devices like smart boards, projectors, and desktop computers, alongside mobile devices like smart phones and laptops. During lectures not all students are always present and even when they are some of the devices demand more bandwidth than others. By using Li-Fi for stationary devices and Wi-Fi for mobile ones, congestion can be significantly reduced and provide a smoother experience for everyone.

Classroom analysis under low, medium and high attendance conditions provides important insights and helped to understand how the network behaves in different real life classroom situations. During low attendance the Wi-Fi network was able to manage the traffic efficiently, as fewer devices were connected and bandwidth demand was moderate. In this case the impact of Li-Fi offloading was present but not very significant because of congestion levels that were already low. However, as attendance increased to medium levels, RF utilization started rising noticeably. More devices meant more data requests and this gradually increased latency and reduced overall network efficiency.

The difference became clearly visible during high attendance. When almost the entire class was present and multiple devices were actively accessing online resources, the Wi-Fi network alone began approaching its maximum utilization capacity. This is where the hybrid model proved its importance. By shifting high bandwidth stationary devices such as the smart board, desktop and projector to Li-Fi reduced the RF load considerably. This redistribution of traffic improved throughput, minimized delay, and ensured that mobile users experienced stable connectivity even during heavy usage periods.

We based our percentage calculations on the estimated number of active devices and their average bandwidth requirements during classroom activities like video streaming, downloading study materials, and collaborative work. Although these values are approximate, they reflect realistic classroom usage patterns and provide a logical justification for our analysis. The reduction in RF congestion was calculated by comparing total devices load before and after offloading stationary devices to Li-Fi. This

method allowed us to clearly demonstrate how the hybrid system enhances performance under different attendance scenarios.

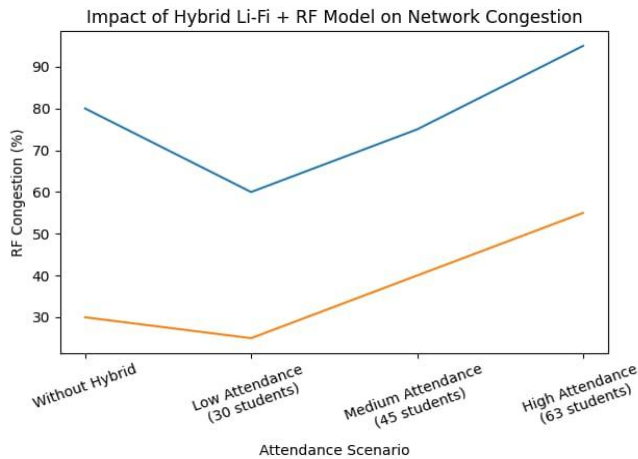


Fig.6. Impact of hybrid model.

Overall, this project shows that integrating RF and Li-Fi technologies is not just a theoretical idea but a practical solution for modern classrooms. It balances mobility and speed by assigning devices according to their nature and bandwidth demand. The system improves reliability, reduces congestion and enhances the digital learning experience for both students and faculty. In conclusion the hybrid RF-Li-Fi approach represents a simple yet effective step toward building smarter, faster, and more efficient classroom networks that are ready to meet future communication demands.

XII. FUTURE SCOPE

Looking at the hybrid RF-Li-Fi system we studied and realized that there are many directions in which it can be extended and improved. One of the first things we thought about was AI based prediction and optimization. Right now, our system detects congestion in real time and offloads traffic to Li-Fi if stationary devices need higher bandwidth. But in the future machine learning models could predict when congestion will happen even before it occurs. For example, the system could learn the typical patterns of classroom usage which lectures are high attendance, when students usually connect multiple devices or which areas of classroom have more mobile device usage and smoothly shifts devices to Li-Fi or adjust Wi-Fi settings to prevent delays. This would make the network smarter and more efficient, minimizing any slowdowns and keeping students' online activities smooth.

Another important direction is the multifloored and larger classroom deployment. Our current analysis is limited to a single classroom, but real campuses have labs, lecture halls, and multifloored buildings. In larger settings, strategically placing multiple Li-Fi access points could cover all stationary devices like projectors, smart boards, and lab computers while still allowing mobile devices to connect via Wi-Fi. This would require a complex network management system but could dramatically improve user experience in bigger learning spaces. It would also be interesting to study how signals from multiple Li-Fi access points could overlap without interference ensuring that all stationary devices always maintain high speed connectivity. We also see potential in integrating internet of thing (IoT)

devices with the hybrid system. Classrooms and labs are increasingly using sensors, automated attendance systems, smart lighting, and environmental monitors. These IoT devices often generate small amounts of data but require constant connectivity. By connecting them intelligently across Li-Fi and Wi-Fi, the network could handle both high speed student devices and low bandwidth IoT devices together. For instance, temperature or air quality could communicate with the smart board or central server through Li-Fi while mobile student devices continue using Wi-Fi keeping both networks efficient.

Another scope is in dynamic handover improvements. In our current model the adaptive controller monitors congestion and decides which stationary devices should be shifted to Li-Fi. But the future system could be even more adaptive, switching users not only based on congestion but also considering device movement, and bandwidth allocation automatically. This would require combining location tracking and adaptive algorithms but could make the hybrid network almost self-managing.

Energy efficiency is also a major area to explore. Since Li-Fi uses LED lighting, combining data transmission with smart lighting could save electricity. For eg LEDs could transmit the data only when the class is in use or dim slightly during low network activity while still maintaining connectivity. Similarly, reducing RF congestion by offloading stationary devices can lower overall power consumption. This could make hybrid networks both environmentally friendly and cost effective for educational institutions.

Finally, security and privacy can be further improved as Li-Fi signals do not penetrate walls, this makes them more secure. But we could enhance security with encryption, authentication layers, and even device specific access controls. For e.g., sensitive administrative data or online exams could use Li-Fi links, while general student Wi-Fi traffic remains separate. This would reduce the risk of data leaks and provide a safe network environment for learning and research. In short, the future of Hybrid RF-Li-Fi networks is full of various possibilities. By combining AI, IoT integration, multifloored deployment, energy optimization, and advanced security, classrooms could become truly intelligent spaces. Students and faculty could experience seamless connectivity, even during high attendance hours, while the network manages itself efficiently. This research opens many doors for experimentation, real world deployment, and innovative improvements that could redefine indoor educational networks.

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